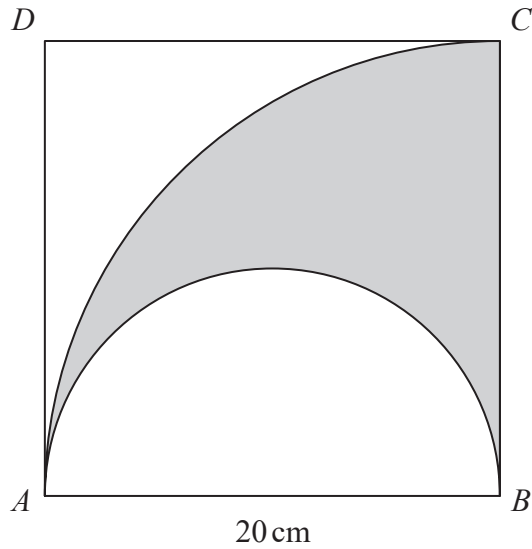


- 7 The diagram shows a square $ABCD$ with sides of length 20 cm. It also shows a semicircle and an arc of a circle.



AB is the diameter of the semicircle.
 AC is an arc of a circle with centre B .

Show that
$$\frac{\text{area of shaded region}}{\text{area of square}} = \frac{\pi}{8}$$